

week_9_lab_tasks

This file geocodes each Airbnb listing to its SA2 area code via the Koordinates API, joins the geocoded Airbnb data with the cleaned tenancy dataset on area and quarter, and answers the three analysis questions (median Central price, biggest short vs long-term rental gap, and Airbnb-vs-rental counts by area).

table id: 123518 stats NZ table url: <https://datafinder.stats.govt.nz/table/123518-geographic-areas-table-2026/> Statistical Area 2 2026, layer id: 123515

https://koopordinates.com/services/query/v1/vector.json?key=KEY&layer=3936&x=172.6398&y=43.5256&max_results=3&radius=20000&geometry=true&with_field_names=true

#Setting up the environment installing packages

Querying the API

Queries the Koordinates API. We give it a longitude (x) and latitude (y) and it sends back data that matches that location. If the request fails for some reason (bad internet, server down, etc) it just returns None

```
## y
## {'vectorQuery': {'layers': {'123515': {'type': 'FeatureCollection', 'crs': {'type': 'name', 'property': 'name', 'value': 'EPSG:4326'}}}}
```

Uses the query function above but pulls out just the area code and the name.

A couple tests

```
## y
```

```
## Ilam University
```

```
## y
```

```
## 326600 Christchurch Central
```

Defining things

This function geocodes all the AirBnB listings at once, excluding duplicates. It looks at every unique lat/long pair, checks if there's already a saved file from a previous run (skips anything already done), then queries the API for whatever's left. Once finished, it merges the area codes back onto the full AirBnB dataset and saves it to CSV.

Loads the cleaned Airbnb CSV and runs it through `run_batch_geocoding()`, which adds the area code and name to every listing based on its coordinates

```
## 148 / 4101 unique coordinates still need querying
## y
## y
## y
```

[illegible]

[illegible]

[illegible]

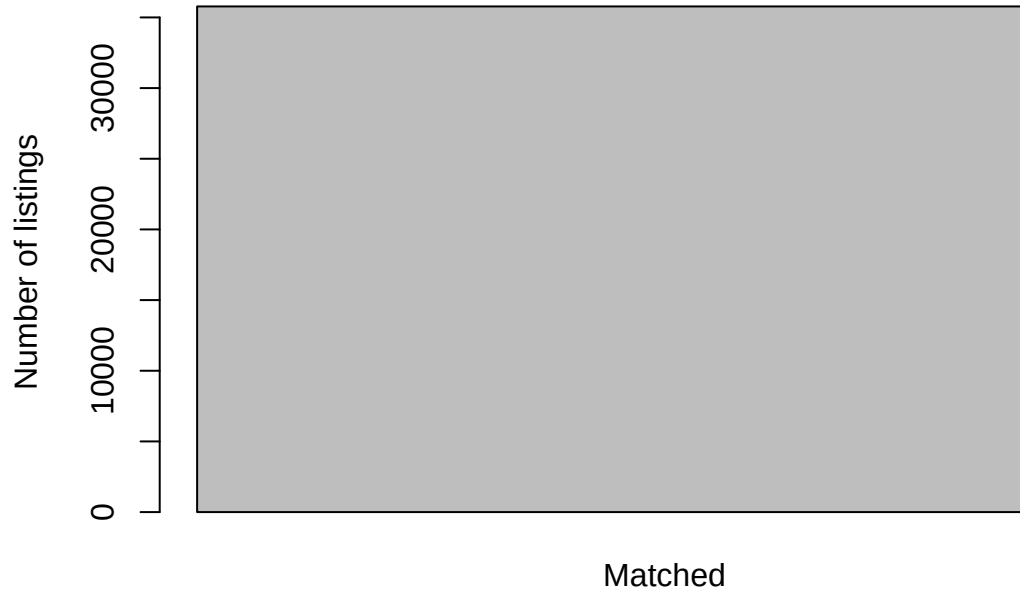
```

##                                     NAs      :10905
## minimum_nights    number_of_reviews  last_review  reviews_per_month
## Min.      : 1.000    Min.      : 0.00    Min.      :2014-04-14    Min.      : 0.000
## 1st Qu.: 1.000    1st Qu.: 5.00    1st Qu.:2025-10-27    1st Qu.: 0.480
## Median : 1.000    Median : 26.00    Median :2026-02-07    Median : 1.450
## Mean   : 1.768    Mean   : 70.51    Mean   :2025-12-01    Mean   : 2.115
## 3rd Qu.: 2.000    3rd Qu.: 85.00    3rd Qu.:2026-04-24    3rd Qu.: 3.050
## Max.   :365.000    Max.   :1601.00    Max.   :2026-08-13    Max.   :16.570
## NAs     :37                                     NAs     :3224
## calculated_host_listings_count  availability_365  number_of_reviews_ltm
## Min.      : 1.00                                     Min.      : 0.0    Min.      : 0
## 1st Qu.: 1.00                                     1st Qu.:109.0    1st Qu.: 2
## Median : 3.00                                     Median :225.0    Median : 10
## Mean   : 57.69                                     Mean   :203.6    Mean   : 19
## 3rd Qu.: 47.00                                     3rd Qu.:301.0    3rd Qu.: 27
## Max.   :1586.00                                     Max.   :365.0    Max.   :185
##
##      month_year      scrape_date      days_since_review      area_code
## Length :35778    Min.      :2025-10-31    Min.      : -2.0    Min.      :316401
## N.unique : 11    1st Qu.:2025-12-31    1st Qu.: 25.0    1st Qu.:324300
## N.blank  : 0    Median :2026-03-31    Median : 38.0    Median :327000
## Min.nchar: 7    Mean   :2026-04-05    Mean   :125.4    Mean   :326627
## Max.nchar: 7    3rd Qu.:2026-06-30    3rd Qu.: 94.0    3rd Qu.:329500
##                                     Max.   :2026-08-31    Max.   :4252.0    Max.   :363800
##                                     NAs     :3224
##
##      area_name
## Length :35778
## N.unique : 178
## N.blank  : 0
## Min.nchar: 4
## Max.nchar: 34
##
##

```

Sanity check to check if any listings failed to geocode (before join): if every listing matched area code on ten-

Sanity check: how many listings got a matched area code



any dataset, there will only be 1 bar

```
## 35778 / 35778 listings matched an area code
```

```
## Rows: 35,778
## Columns: 20
## $ id          <dbl> 319770, 737452, 1004252, 1007247, 10974~
## $ name        <chr> "Gateway to the South Island", "Mumford~
## $ host_id     <dbl> 1639584, 3851282, 5523780, 4765250, 499~
## $ neighbourhood <chr> "Innes Ward", "Banks Peninsula Ward", "~
## $ latitude    <dbl> -43.50961, -43.82343, -43.61889, -43.53~
## $ longitude   <dbl> 172.6555, 172.9464, 172.6531, 172.6070,~
## $ room_type   <chr> "Private room", "Entire home/apt", "Ent~
## $ price       <dbl> 196, 394, 183, 305, 189, 69, 337, 166, ~
## $ minimum_nights <dbl> 1, 2, 2, 2, 1, 5, 3, 2, 2, 2, 1, 2, 2, ~
## $ number_of_reviews <dbl> 3, 102, 227, 119, 230, 205, 16, 181, 98~
## $ last_review  <date> 2015-01-09, 2026-04-11, 2026-04-06, 20~
## $ reviews_per_month <dbl> 0.02, 0.68, 1.44, 0.79, 2.05, 1.33, 0.1~
## $ calculated_host_listings_count <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 3, 1, 1, ~
## $ availability_365 <dbl> 364, 194, 173, 304, 295, 309, 45, 344, ~
## $ number_of_reviews_ltm <dbl> 0, 3, 10, 5, 0, 23, 0, 11, 15, 6, 54, 7~
## $ month_year    <chr> "2026-04", "2026-04", "2026-04", "2026--~
## $ scrape_date   <date> 2026-04-30, 2026-04-30, 2026-04-30, 20~
## $ days_since_review <dbl> 4129, 19, 24, 68, 423, 11, 696, 68, 72,~
## $ area_code     <dbl> 325900, 333300, 332200, 325500, 322200,~
## $ area_name     <chr> "Richmond North (Christchurch City)", "~
```

Joins the geocoded Airbnb data with the tenancy bond data on area_code and quarter, so each Airbnb

listing gets matched with the long-term rental stats for its area and time period. It first converts Airbnb's monthly dates into quarters and makes sure area_code is numeric (to match tenancy's format), then filters the tenancy data down to just the overall totals (Dwelling Type = "ALL", Number Of Beds = "ALL") and drops rows with an invalid location ID (-99). A left join is used so every Airbnb listing is kept, even if there's no matching tenancy data for that area/quarter. The result is saved as airbnb_tenancy_joined.csv.

Sanity check - a left_join can silently multiply rows if the right-hand table (tenancy_all) has more than one row per join key. We already filtered tenancy_all to one row per (area_code, quarter) above, but asserting it here means we'll get an immediate, obvious error if that ever stops being true (e.g. after someone edits the filter above), instead of a silently-inflated dataset

```
## Sanity check passed: joined has the same number of rows as airbnb ( 35778 rows ) - the join did not c
```

```
## Rows: 35,778
## Columns: 33
## $ id          <dbl> 319770, 737452, 1004252, 1007247, 10974~
## $ name        <chr> "Gateway to the South Island", "Mumford~
## $ host_id     <dbl> 1639584, 3851282, 5523780, 4765250, 499~
## $ neighbourhood <chr> "Innes Ward", "Banks Peninsula Ward", "~
## $ latitude    <dbl> -43.50961, -43.82343, -43.61889, -43.53~
## $ longitude   <dbl> 172.6555, 172.9464, 172.6531, 172.6070,~
## $ room_type   <chr> "Private room", "Entire home/apt", "Ent~
## $ price       <dbl> 196, 394, 183, 305, 189, 69, 337, 166, ~
## $ minimum_nights <dbl> 1, 2, 2, 2, 1, 5, 3, 2, 2, 2, 1, 2, 2, ~
## $ number_of_reviews <dbl> 3, 102, 227, 119, 230, 205, 16, 181, 98~
## $ last_review  <date> 2015-01-09, 2026-04-11, 2026-04-06, 20~
## $ reviews_per_month <dbl> 0.02, 0.68, 1.44, 0.79, 2.05, 1.33, 0.1~
## $ calculated_host_listings_count <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 3, 1, 1, ~
## $ availability_365 <dbl> 364, 194, 173, 304, 295, 309, 45, 344, ~
## $ number_of_reviews_ltm <dbl> 0, 3, 10, 5, 0, 23, 0, 11, 15, 6, 54, 7~
## $ month_year   <chr> "2026-04", "2026-04", "2026-04", "2026--
## $ scrape_date  <date> 2026-04-30, 2026-04-30, 2026-04-30, 20~
## $ days_since_review <dbl> 4129, 19, 24, 68, 423, 11, 696, 68, 72,~
## $ area_code    <dbl> 325900, 333300, 332200, 325500, 322200,~
## $ area_name    <chr> "Richmond North (Christchurch City)", "~
## $ month_date   <date> 2026-04-01, 2026-04-01, 2026-04-01, 20~
## $ quarter      <date> 2026-04-01, 2026-04-01, 2026-04-01, 20~
## $ TimeFrame    <date> 2026-04-01, NA, NA, NA, 2026-04-01, 20~
## $ 'Dwelling Type' <chr> "ALL", NA, NA, NA, "ALL", "ALL", "ALL",~
## $ 'Number Of Beds' <chr> "ALL", NA, NA, NA, "ALL", "ALL", "ALL",~
## $ 'Total Bonds' <dbl> 21, NA, NA, NA, 33, 9, 15, NA, 12, NA, ~
## $ 'Active Bonds' <dbl> 342, NA, NA, NA, 270, 147, 153, NA, 234~
## $ 'Closed Bonds' <dbl> 15, NA, NA, NA, 24, 9, 15, NA, 12, NA, ~
## $ 'Median Rent' <dbl> 530, NA, NA, NA, 565, 650, 700, NA, 640~
## $ 'Geometric Mean Rent' <dbl> 539, NA, NA, NA, 567, 572, 689, NA, 677~
## $ 'Upper Quartile Rent' <dbl> 640, NA, NA, NA, 723, 698, 800, NA, 685~
## $ 'Lower Quartile Rent' <dbl> 510, NA, NA, NA, 550, 578, 610, NA, 601~
## $ 'Log Std Dev Weekly Rent' <dbl> 0.2668, NA, NA, NA, 0.4410, 0.3762, 0.3~
```

17/9/2026 - Alice • Dropped rows with location ID -99 from the Week 8 tenancy file and filtered to keep only rows where Dwelling Type == "ALL" and Number Of Beds == "ALL". (This makes sure there is only 1 row per suburb and quarter, which prevents our Airbnb listings from getting duplicated when we merge the files.)

- Created week_9_datasets folder, which contains the two cleaned Week 8 files: Airbnb_cleaned_week_8 and tenancy_cleaned_week_8.
- Airbnb data has latitude/longitude, while tenancy data has area codes. Used the Koordinates API to geocode each unique Airbnb coordinate to its official SA2 area code.
- Downloaded geographic-areas-tables-2026.csv (originally a zip file with 5 files in it, including geographic-areas-table-2026.csv) from Koordinates
- Created run_batch_geocoding(), which uses Alisha's get_area_info() and query_layer() functions. It batch-geocoded ~4,000 unique coordinates. The function saves the result as airbnb_with_area_codes.csv with a new area_code column. It also checks for existing output and only queries coordinates with missing area codes, avoiding unnecessary API calls from now on.
- Joined the geocoded Airbnb data with tenancy_cleaned_week_8 using a left join, keeping all Airbnb listings. It first converted Airbnb's monthly dates into quarterly periods to match the rental data, then joined on both area_code and quarter. Used a left join so Airbnb listings are retained even when there is no matching rental data for their area/quarter. (Inner join will delete any data without matching rental data. Full/outer join will include tenancy rows that are NOT in chch, which is not meaningful)
- Saved the final combined dataset as airbnb_tenancy_joined.csv.

Note merged data assigned variable name: joined_data Heads up airbnb_tenancy_joined.csv is the finished joined data: joined_data <- read_csv('week_9_datasets/airbnb_tenancy_joined.csv'. So mess with joined_data however you want for your part. Just not write _csv() back to that exact filename (hopefully we won't have to write another csv file but just in case) otherwise will overwrite it for everyone. If we need to save our own version, give it a totally different name, like price_gap.csv or something so it doesn't get confused with the original.

What's the median AirBnB price in Christchurch Central (Location ID 326600) - Alice

```
## num [1:35778] 325900 333300 332200 325500 322200 ...

## # A tibble: 6 x 33
##       id name          host_id neighbourhood latitude longitude room_type price
##       <dbl> <chr>          <dbl> <chr>          <dbl>    <dbl> <chr>    <dbl>
## 1 18899315 CBD Loft St~  9.89e7 Central Ward    -43.5    173. Entire h~   520
## 2 22656416 RELAX IN TH~  1.67e8 Central Ward    -43.5    173. Entire h~   256
## 3 24182612 Cityscape a~  1.82e8 Central Ward    -43.5    173. Entire h~   315
## 4 27201423 AMAZING HIS~  1.67e8 Central Ward    -43.5    173. Entire h~   250
## 5 29661763 Luxury/Bran~  5.68e7 Central Ward    -43.5    173. Entire h~   272
## 6 29661813 Nestle Stud~  5.68e7 Central Ward    -43.5    173. Entire h~   204
## # i 25 more variables: minimum_nights <dbl>, number_of_reviews <dbl>,
## #   last_review <date>, reviews_per_month <dbl>,
## #   calculated_host_listings_count <dbl>, availability_365 <dbl>,
## #   number_of_reviews_ltm <dbl>, month_year <chr>, scrape_date <date>,
## #   days_since_review <dbl>, area_code <dbl>, area_name <chr>,
## #   month_date <date>, quarter <date>, TimeFrame <date>, 'Dwelling Type' <chr>,
## #   'Number Of Beds' <chr>, 'Total Bonds' <dbl>, 'Active Bonds' <dbl>, ...

## [1] 957

## [1] 245
```

```
## [1] 210
```

Summary 21/9/2026 - Alice Median airbnb price in CHCH central (area code 326600) is \$239 (based on 724 listings with a recorded price)

In which part of Christchurch can we observe the craziest (largest) gap between short- and long-term rental prices? - Alisha

```
## Rows: 35,778
## Columns: 33
## $ id                <dbl> 319770, 737452, 1004252, 1007247, 10974~
## $ name              <chr> "Gateway to the South Island", "Mumford~
## $ host_id          <dbl> 1639584, 3851282, 5523780, 4765250, 499~
## $ neighbourhood    <chr> "Innes Ward", "Banks Peninsula Ward", "~
## $ latitude         <dbl> -43.50961, -43.82343, -43.61889, -43.53~
## $ longitude        <dbl> 172.6555, 172.9464, 172.6531, 172.6070,~
## $ room_type        <chr> "Private room", "Entire home/apt", "Ent~
## $ price            <dbl> 196, 394, 183, 305, 189, 69, 337, 166, ~
## $ minimum_nights   <dbl> 1, 2, 2, 2, 1, 5, 3, 2, 2, 1, 2, 2, ~
## $ number_of_reviews <dbl> 3, 102, 227, 119, 230, 205, 16, 181, 98~
## $ last_review      <date> 2015-01-09, 2026-04-11, 2026-04-06, 20~
## $ reviews_per_month <dbl> 0.02, 0.68, 1.44, 0.79, 2.05, 1.33, 0.1~
## $ calculated_host_listings_count <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 3, 1, 1, ~
## $ availability_365 <dbl> 364, 194, 173, 304, 295, 309, 45, 344, ~
## $ number_of_reviews_ltm <dbl> 0, 3, 10, 5, 0, 23, 0, 11, 15, 6, 54, 7~
## $ month_year       <chr> "2026-04", "2026-04", "2026-04", "2026--
## $ scrape_date      <date> 2026-04-30, 2026-04-30, 2026-04-30, 20~
## $ days_since_review <dbl> 4129, 19, 24, 68, 423, 11, 696, 68, 72,~
## $ area_code        <dbl> 325900, 333300, 332200, 325500, 322200,~
## $ area_name        <chr> "Richmond North (Christchurch City)", "~
## $ month_date       <date> 2026-04-01, 2026-04-01, 2026-04-01, 20~
## $ quarter         <date> 2026-04-01, 2026-04-01, 2026-04-01, 20~
## $ TimeFrame       <date> 2026-04-01, NA, NA, NA, 2026-04-01, 20~
## $ 'Dwelling Type' <chr> "ALL", NA, NA, NA, "ALL", "ALL", "ALL",~
## $ 'Number Of Beds' <chr> "ALL", NA, NA, NA, "ALL", "ALL", "ALL",~
## $ 'Total Bonds'   <dbl> 21, NA, NA, NA, 33, 9, 15, NA, 12, NA, ~
## $ 'Active Bonds'  <dbl> 342, NA, NA, NA, 270, 147, 153, NA, 234~
## $ 'Closed Bonds'  <dbl> 15, NA, NA, NA, 24, 9, 15, NA, 12, NA, ~
## $ 'Median Rent'   <dbl> 530, NA, NA, NA, 565, 650, 700, NA, 640~
## $ 'Geometric Mean Rent' <dbl> 539, NA, NA, NA, 567, 572, 689, NA, 677~
## $ 'Upper Quartile Rent' <dbl> 640, NA, NA, NA, 723, 698, 800, NA, 685~
## $ 'Lower Quartile Rent' <dbl> 510, NA, NA, NA, 550, 578, 610, NA, 601~
## $ 'Log Std Dev Weekly Rent' <dbl> 0.2668, NA, NA, NA, 0.4410, 0.3762, 0.3~
```

Adding price_gap and filtering data

suberb analysis (SA2)

Most areas have between 1 and 2000 properties. This makes sense as some areas have much more air bnb listings than others. This makes sense as areas like the city will be much more likely to have listings than smaller futher out areas like Aranui.

Decided to only include areas with over 15 properties listed

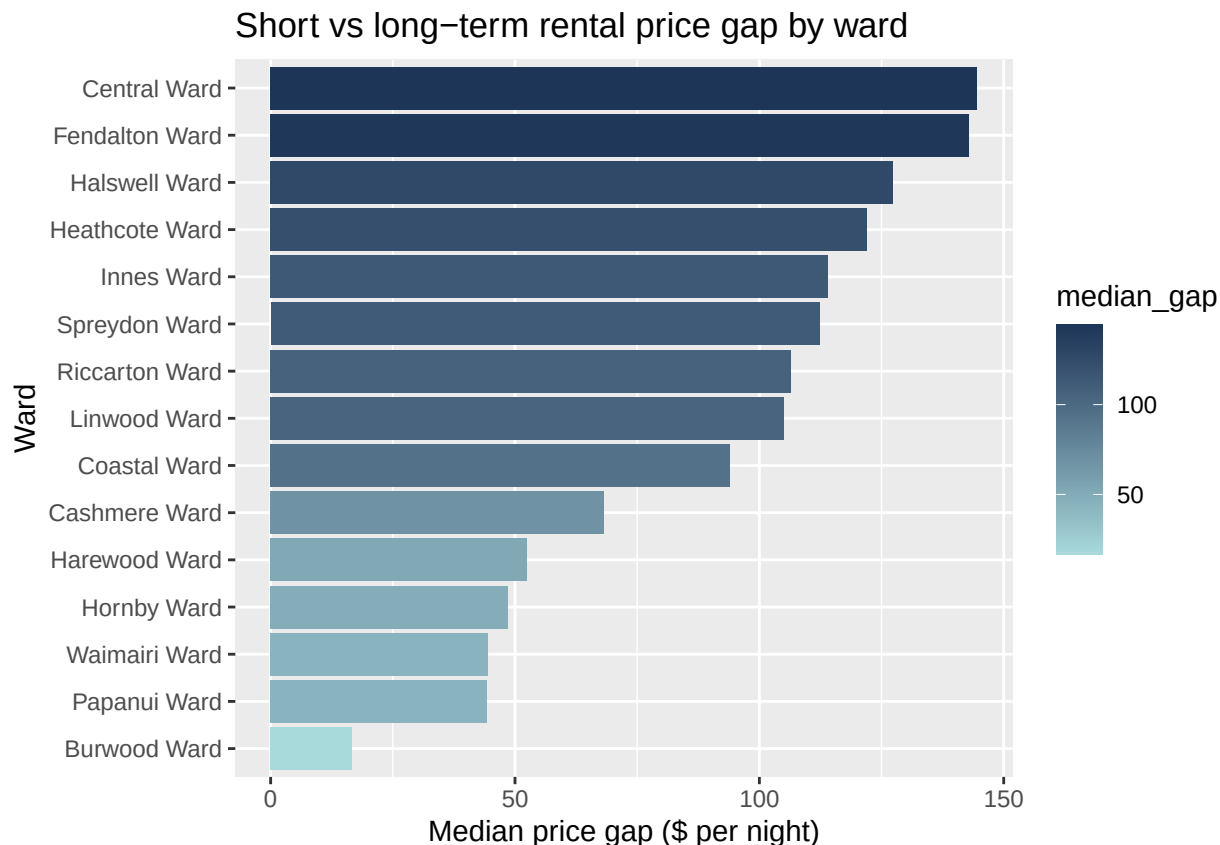
```
## # A tibble: 15 x 4
##   area_code area_name median_gap n
##   <dbl> <chr>      <dbl> <int>
## 1 322600 Holmwood      216.    73
## 2 317400 Northwood     204.    28
## 3 332700 Sumner       199.   351
## 4 322100 Malvern      195.    86
## 5 322500 Wigram North  188.    44
## 6 330500 Ensors       187.    41
## 7 324100 Wigram East   183.    29
## 8 332100 Redcliffs     175.   153
## 9 325700 Christchurch Central-West 175.   735
## 10 319400 Papanui North 171.    20
## 11 327700 Hoon Hay East 167.    32
## 12 323600 Wigram South 166.    26
## 13 326800 Richmond South (Christchurch City) 162.   120
## 14 326600 Christchurch Central 161.   719
## 15 327000 Christchurch Central-East 158.  1782
```

Even after filtering, some SA2 suburb areas (like Holmwood) are small, obscure statistical areas rather than widely-known suburbs. Tried grouping by ward instead because wards are bigger areas with higher listing counts and are more recognisable:

Ward-level analysis

```
## # A tibble: 15 x 3
##   neighbourhood median_gap n
##   <chr>      <dbl> <int>
## 1 Central Ward      145.  6410
## 2 Fendalton Ward    143.  1002
## 3 Halswell Ward     127.   345
## 4 Heathcote Ward    122.  1210
## 5 Innes Ward        114.   483
## 6 Spreydon Ward     112.   617
## 7 Riccarton Ward    106.  1116
## 8 Linwood Ward      105.   422
## 9 Coastal Ward       94.   501
## 10 Cashmere Ward     68.1   588
## 11 Harewood Ward     52.4   343
## 12 Hornby Ward       48.6   290
## 13 Waimairi Ward     44.4   410
## 14 Papanui Ward      44.3   309
## 15 Burwood Ward      16.7   110
```

Looking at the wards they look like more well known official areas of Christchurch



«««< **HEAD Answer:** At the ward level (larger, more reliable sample sizes), Central Ward shows the largest gap between short- and long-term rental prices at \$144.57/night (n=6426).

(At the smaller SA2 suburb level, Holmwood shows an even larger gap (\$217.21/night, n=74), but it's a small statistical area rather than a widely-recognised suburb.)

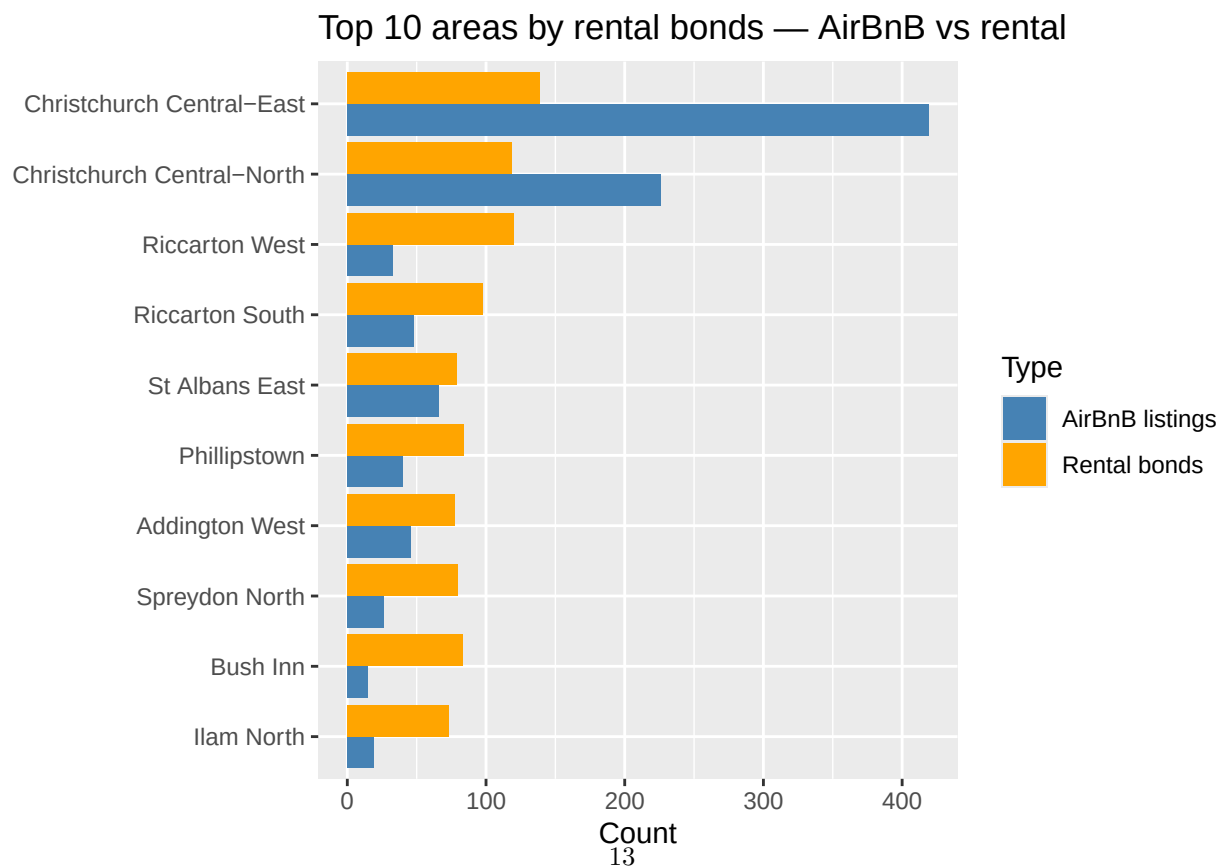
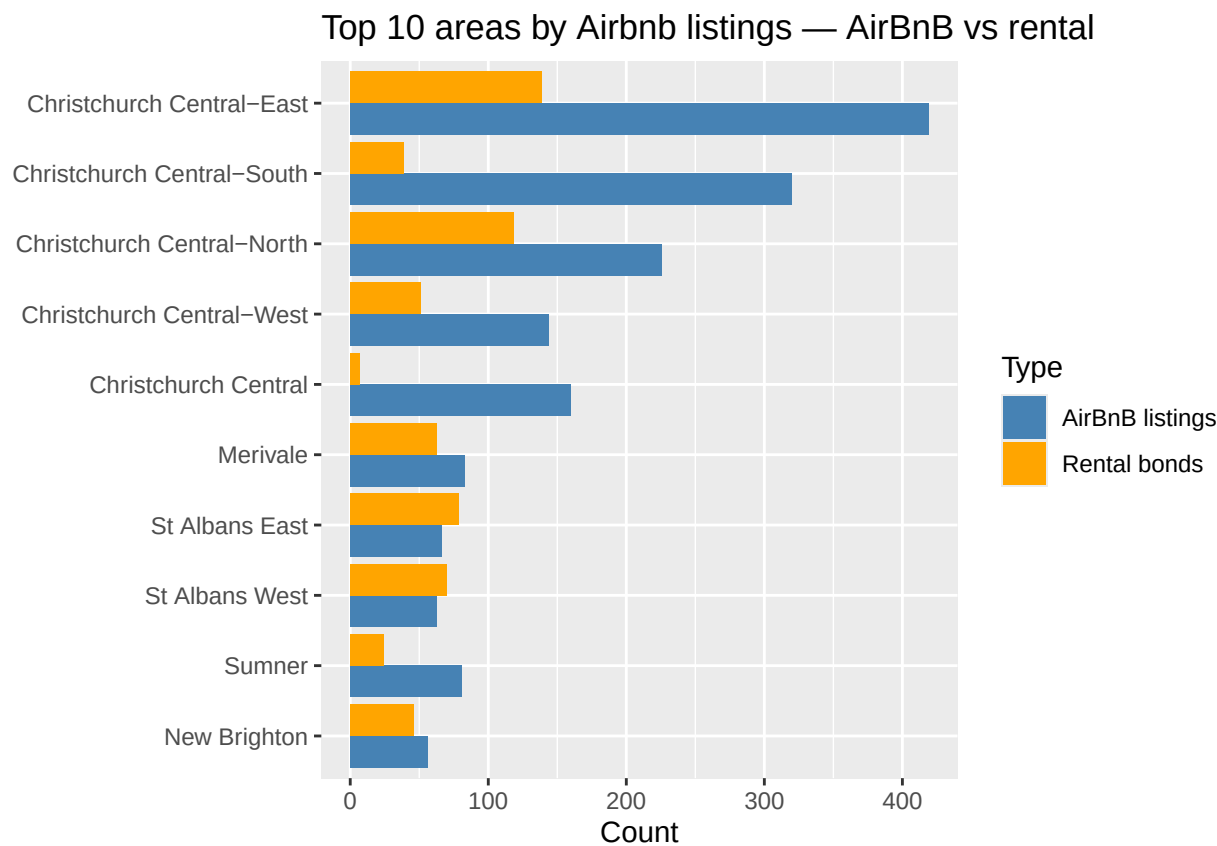
The ward with the greatest gap between short- and long-term rental prices is Central Ward (the CBD), with a median gap of \$144.57 per night (n=6426). »»»> a46693758be552d0f70c3a07eb191f0d0c8f53e6

Compare how many AirBnBs and rental properties we have in each location - Charlie

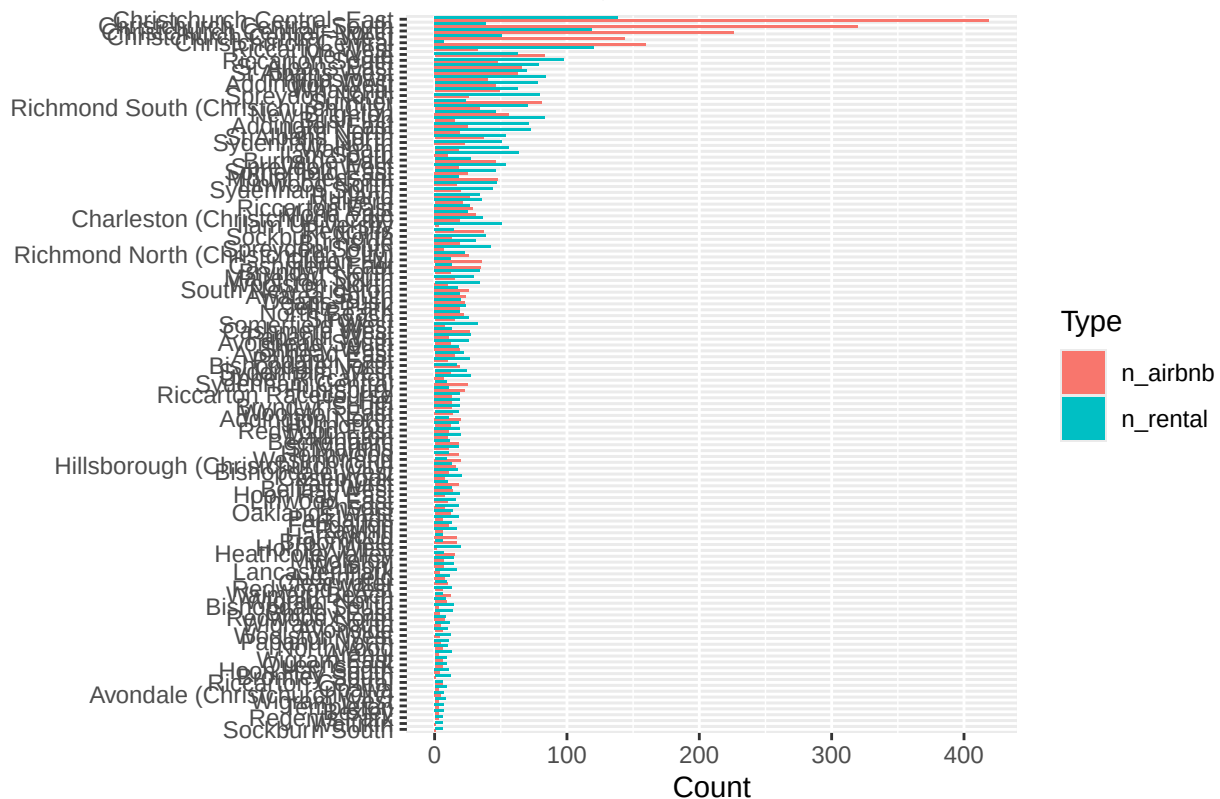
```
## # A tibble: 178 x 4
##   area_code area_name          n_airbnb n_rental
##   <dbl> <chr>              <int>    <dbl>
## 1  327000 Christchurch Central-East    419    139.
## 2  327100 Christchurch Central-South   320     39.1
## 3  325800 Christchurch Central-North   226    119.
## 4  326600 Christchurch Central         160      7.07
## 5  325700 Christchurch Central-West    144    51.0
## 6  333500 Akaroa                    120     NaN
```

##	7	333300 Akaroa Harbour	95	NaN
##	8	323000 Merivale	83	62.8
##	9	332700 Sumner	81	24.0
##	10	332501 Lyttelton	72	NaN
##	# i	168 more rows		

filter NA bonds as these are outside of the tenancy nz scope



AirBnB listings vs rental bonds — all areas



Summary In the top 10 Airbnb listings we can see that central Christchurch dominates by a large margin. The central Christchurch suburbs make up 5 out of the top 10, with central east having the most at around 400 listings. The graph also shows us how much the central suburbs have compared to the rest. There are 1563 Airbnb listings spread across the top 10 suburbs of which central suburbs make up 1128. Central suburbs also have very few rental bonds compared to Airbnb's.

The top 10 rental bond suburbs shows different areas. Although it is led by central east and north I predict this is mostly due to the larger population and apartment buildings in this area. The rest of the top 10 are full of residential areas. The more interesting part is how the all of these areas have significantly more rental bonds than Airbnb's.

The last graph shows us how Airbnb's very significantly dominates the most populated areas, we can see the massive spike at the top in central suburbs. After this initial spike in the central suburbs there are almost none that have more Airbnb listings than rental bonds.

Overall we can conclude that the Airbnb market and long term rental market are concentrated in different areas, other than central east and north which have large volumes of both. Airbnb significantly dominates central suburbs with long term rent has greater volumes almost everywhere else.

natural language description:

1. Set up the tools We got an API key from Koordinates and found the dataset we needed — Statistical Area 2 (SA2) boundaries for 2026.
2. Built a way to turn coordinates into area codes Our Airbnb data only had latitude/longitude for each listing, but our rental bond data used area codes instead. We wrote a function (`query_layer`) that sends a coordinate to the Koordinates API and gets back the area that the point falls inside, then a second function (`get_area_info`) that pulls out just the area code and name of the area.

3. Ran this across the whole Airbnb dataset Since some listings share the exact same coordinates (duplicate buildings/addresses), we only queried the API for unique coordinates rather than every single row. We also made sure that if we ever needed to re-run this, it would skip anything already done instead of starting from scratch. This function ran through multiple requests in parallel (multithreading) and saved the results to a CSV so we'd never have to redo it.
4. Joined the two datasets together With area codes attached to each Airbnb listing, we matched on area code and time — converting Airbnb's monthly data into quarters (since the rental data is reported quarterly). We used a left join so every Airbnb listing was kept, even if there wasn't a matching rental record for that area/quarter.
5. Answered the three main questions: Median Airbnb price in Christchurch Central: Filtered to just that area code and calculated the median — came out to \$239/night, based on 724 listings.

Where's the biggest gap between short-term and long-term rental prices: We calculated the difference between Airbnb's nightly price and the long-term rental's price (converted to nightly), then compared this across different areas. At the small suburb (SA2) level, Holmwood had the biggest gap, but most of the SA's were rather small and not well known. So we also grouped by wards and found that the Central Ward had the largest gap (\$144.57/night).

Comparing Airbnb vs rental property counts by area: Counted how many Airbnb listings compared to how many rental bonds existed in each area. We found that Airbnb listings are heavily concentrated in central Christchurch, while long-term rentals are much more spread out across the wider city — meaning the two markets are mostly operating in different areas, except in the central suburbs where both overlap.